Speed Dating Data: Exploratory Data Analysis

Kacper Kuźnik · Aleksander Hanski · Piotr Masłowski

1 Data

The data come from a speed-dating experiment run at Columbia Business School in 2002–2004 (Kaggle). Participants, mostly graduate students, filled in a sign-up survey and attended one of 21 events (*waves*, 9–44 people each). At an event everyone had a four-minute date with every participant of the opposite sex and then said *yes* or *no*; a **match** means both said yes. We model this as a graph: people are nodes, dates are edges, and the task is to predict which edges are matches.

people (nodes)	544 (269 women, 275 men)	dates per person	4–22 (median 16)
dates (edges)	4,105	matches per person	0–14 (median 2)
matches	675 (16.4% of dates)	match rate per event	8%–31%

Table 1: The graph after preprocessing.

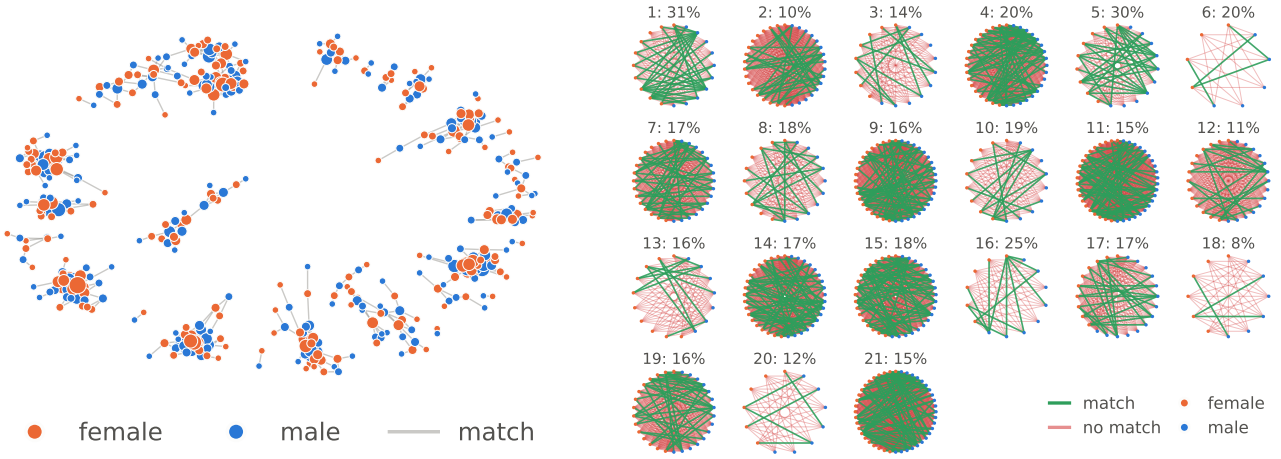
2 Preprocessing

The raw file (8,378 rows × 195 columns) has one row per person per date, and its column names are codes that must be looked up in the codebook (e.g. `attr1_1` = how much a person values *attractive* at sign-up). We keep only the 40 sign-up survey columns, since anything recorded during or after a date would leak the outcome, and build one profile per person and one row per date. Small fixes: hobby scores are clipped to 0–10, the six "what do you look for" answers (a 100-point split) are turned into shares that add up to 1, and codes become labels. Seven people who skipped practically the whole survey (34–37 of 37 answers missing) are dropped with their 79 dates, as the model would see them only as imputed medians. In the remaining profiles 536 of 544 people answered everything; the other 8 miss 1–5 answers, which training fills with the median.

3 Exploratory analysis

3.1 The graph and the participants

People only dated others at their own event, so the graph splits into 21 disconnected clusters (Fig. 1a). Within an event the dates form a complete bipartite graph: everyone met everyone of the opposite sex (Fig. 1b). Matches are sparse: the median person has 2, and 102 people (19%) have none. The match rate also varies a lot between events, from 8% to 31%.



(a) Match network. Each cluster is one event; node size = number of matches. The 102 people without a match are not shown.

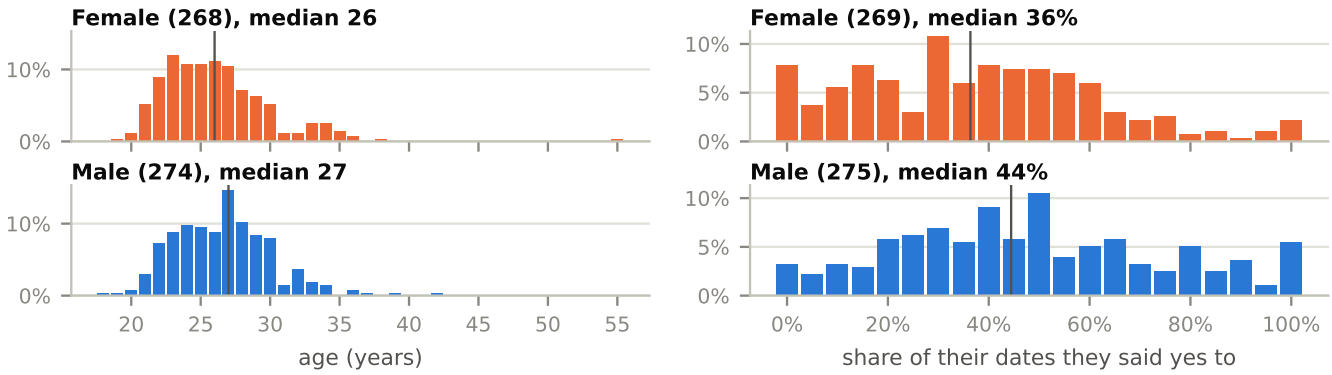
(b) Every date, one circle per event (label: event and its match rate). Women on the left half, men on the right; each line is one date.

Figure 1: The match graph and all dates per event.

Most participants are 21–30 (88%; median age 26 for women, 27 for men; Fig. 2a). Both genders came mainly for a fun night out (about 42%) or to meet new people (32–37%); only 5% of women and 10% of men came "to get a date", and 3–5% for a serious relationship. Fields of study differ strongly: 36% of men study business/economics and 16% engineering, while women are spread evenly over social science, education, natural sciences, business and social work (11–13% each).

3.2 Stated vs revealed preferences

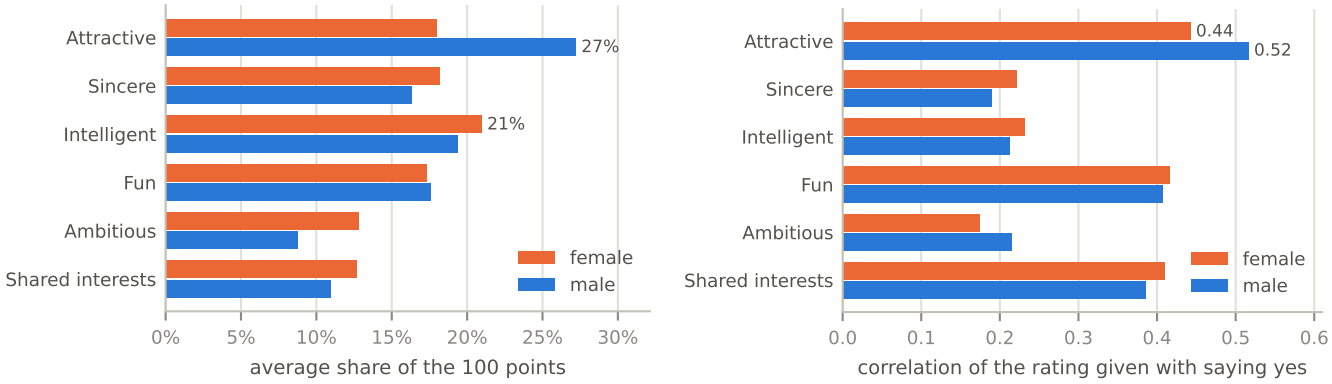
Asked at sign-up, men put attractiveness far ahead (27% of their points), while women spread their points more evenly with intelligence first (21%) and value ambition and shared interests more than men (Fig. 3a). Both sexes overestimate how much the other side cares about looks: women think men give 36% to attractiveness, men think women give 25% (vs 27% and 18% actually stated). Decisions tell a different story (Fig. 3b): for *both* genders the



(a) Age, as the share of each gender per year.

(b) Share of their dates each person said yes to.

Figure 2: Age of the participants and how selective they were.



(a) *Stated*: average share of points given to each attribute at sign-up.

(b) *Revealed*: correlation between the rating given to a partner and saying yes.

Figure 3: What people say they look for vs what goes with a yes.

rating of attractiveness is the attribute most associated with a yes (women 0.44, men 0.52), followed by fun and shared interests (about 0.4), while intelligence and sincerity, which women rank first when asked, matter much less (about 0.2).

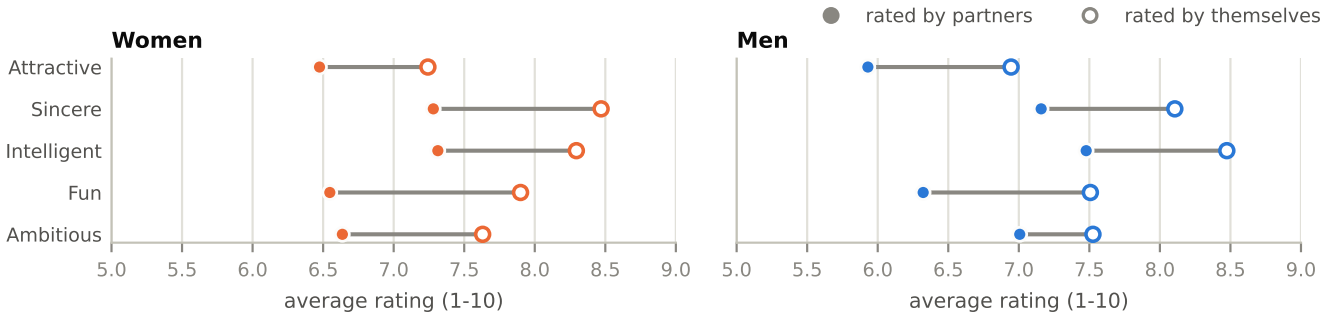


Figure 4: Self-rating at sign-up (hollow) vs the average rating received from date partners (filled), 1–10.

3.3 Self-image and pickiness

Everyone rates themselves higher than their partners did, on every attribute and mostly by about one point (Fig. 4). The largest gap is for *fun* (women 7.9 vs 6.5, men 7.5 vs 6.3), and men rate their own looks a full point above what their dates gave them (7.0 vs 5.9). Men are also less selective (Fig. 2b): the median man said yes to 44% of his dates, the median woman to 36%. Both distributions are wide; 21 women and 9 men said yes to nobody, 6 women and 15 men to everyone.

4 Implications for the model

- **Class imbalance.** Only 16.4% of dates are matches.
- **Graph structure.** Events are disconnected and many people have 0–2 matches.

These are possibly some obstacles, but we will overcome them!